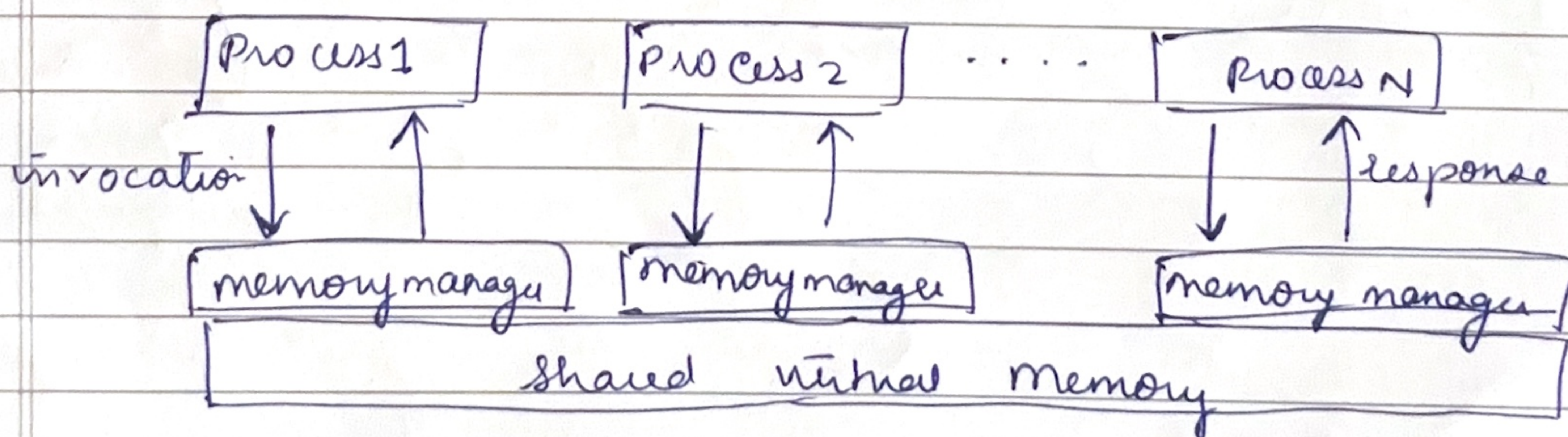


Experiment No 9

Distributed Shared Memory

What is distributed shared memory?

- It is mechanism that manages memory across multiple nodes and makes inter-process communication transparent to end users.
- The application will think that they are running ^{on a} process. Shared memory DSM is a mechanism of allowing use processes to access shared data without using interprocess communications. In DSM every node has its own memory and provides memory read and write services and it provides consistency protocols.
- The distributed shared memory (DSM) implements the shared memory model in distributed systems but it doesn't have physically shared memory. All the nodes share the virtual address space provided by the shared memory model. The data moves between the main memory of different nodes.



Types of distributed shared memory

1. On-chip memory

- The data is present in the CPU position
- Memory is directly connected to address line
- On chip memory DSM is expensive

2. Bus-Based Multiprocessors.

- A set of parallel wires called a bus acts as a connection
- Accessing of same memory simultaneously by multiple CPU's is prevented by using some algorithms
- Cache memory is used to reduce network traffic

3. Ring Based Multiprocessors.

- A set of parallel wires called a bus acts as a connection between CPU and memory.
- Accessing of same memory simultaneously by multiple CPU's is prevented by using some algorithms.

Advantages of Distributed Shared Memory.

- Data sharing is implicit, hiding data movement
- passing data structures containing pointers is easier (in message passing model data moves between different address space)
- Moving entire object to user takes advantage of locality difference
- less expensive to build than tightly coupled Multiprocessor system.

Algorithm for implementing DSM.

① The central server algorithm:

- Central Server maintains all shared data
- Implementation
 - A time out is used to resend request if ack fails.
 - Associated Sequence numbers can be used to detect duplicate write request
 - If an application request to access shared data fails repeatedly, a failure condition is sent to application.

② The migration algorithm - operation

- Ship (migrate) entire data object (page, block) containing data item to requesting location
- allow only one node to access a shared data at a time

③ Read Replication algorithm.

- Replicates data objects to multiple nodes
- DSM keeps track of location of data objects.
- Multiple nodes can have read access or one node write access (multiple readers - one write protocol).
- After a write, all copies are invalidated or updated.

④ Full replication model.

- Extension of read application: multiple nodes can read and multiple nodes can write (multiple readers and multiple writers protocol).

Experiment No 10

Distributed file System (AFS/CODA).

Theory:

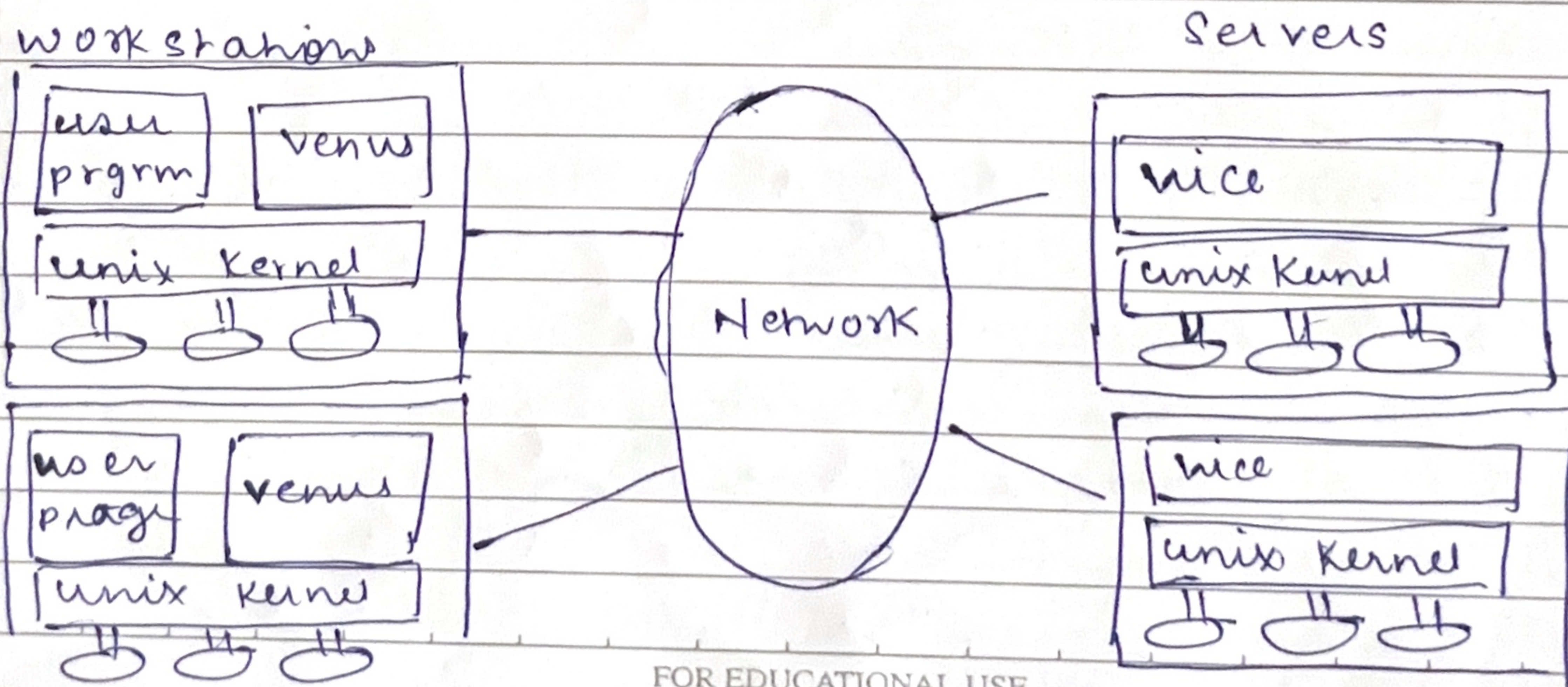
- Andrew file System (AFS) is a distributed file system that allows multiple computers to share file and data seamlessly. It was developed by morris Et al. in 1986 at Carnegie Mellon University in collaboration with IBM.
- AFS was designed to make it easier for people working on different computers to access and share files as if they are all on the same machine, providing a way to manage files efficiently across a network.
- After logging on to workstations that communicate inside the Distributed Computing Infrastructure users exchange data and program (DCI).
- The goal is to facilitate large-scale information exchange by reducing client-server communication. This is accomplished by moving whole files between server and client computer and caching them until the servers get a more recent version.

Andrew file System Architecture

⇒ **nice**: The AFS provides a homogenous, location transparent file namespace to all client workstations by utilizing a group of trustworthy servers known as **nice**.

• The Berkeley Software Distribution of the Unix operating system is used on both clients and server. Each workstation operating system intercepts file system calls and redirects them to a user-level process on that workstation.

⇒ **venus**: This mechanism, known as **venus**, caches files from **nice** and returns updated version of those files to the servers from which they originated. Only when a file is opened or closed does **venus** communicate with **nice**: individual bytes of the cached copy skipping **venus**.



The following are the server and client components used in AFS

- Any Computer that creates requests for AFS server files hosted on a network qualifies as a client.
- The file is saved in the client machine's local cache and shown to the user once a server responds and transmits a requested file.
- When a user visits the AFS, the client sends all modifications to the server via a callback mechanism. The client machine's local cache stores frequently used files for rapid access.

Advantages.

- Shared files that aren't updated very often and local user files that aren't updated too often will last a long time.
- It sets up a lot of storage space for caching.
- It offers a big enough working set for all of a user's files, ensuring that the file is still in the cache when the user accesses it again.